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TIA Add-In: PN Watchdog

Central management of PROFINET watchdog factors in the TIA Portal

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1. Introduction

1.1. Overview

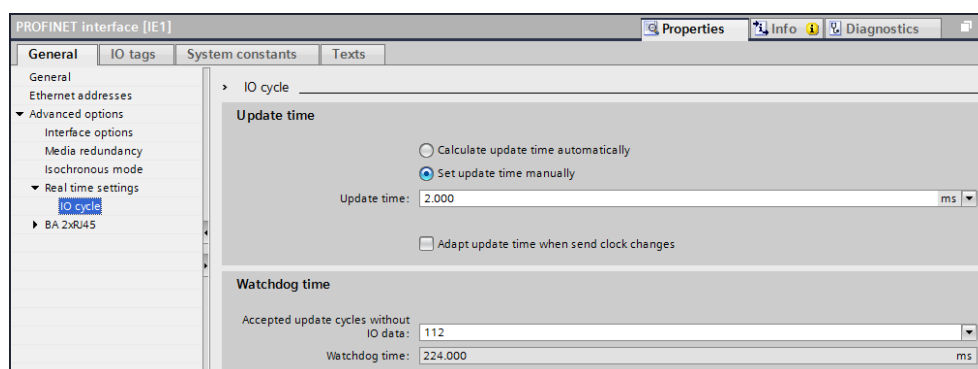
The PN Watchdog Add-In is used for the central management of the factor "Accepted update cycles without IO data" for one, several or all IO systems of a TIA Portal project. Instead of having to go to the effort of setting this watchdog factor for each PN IO device individually in the TIA Portal, this Add-In offers a clearly structured view in which the factor setting can be made with just a few clicks.

The PROFINET Watchdog Add-In is a functional extension of the R/H Add-In and fully replaces this. The primary objective is MRP configurations in which the "Accepted update cycles without IO data" factor must be increased due to the expected MRP reconfiguration time. Below, we refer to this factor simply as the watchdog factor.

In redundant S7-1500 systems, a minimum value is mandatory if the redundancy PLCs are nodes in the ring.

In all other systems, a watchdog time of 200 ms is recommended for MRP configurations because it can be guaranteed that an MRP ring is reconfigured after this time.

Figure 1-1: IO cycle properties of a PN IO device



1.2. Mode of operation

The Add-In offers the "Show all PN IO-Devices" functions via the shortcut menu. All PN IO devices connected to the selected systems are listed. Grouping takes place by controller system and then by interface.

In the list, you can set an individual factor or let the tool make the corresponding setting for each device.

Redundancy system with MRP

In redundant systems of the S7-1500 in which the redundant controllers are ring nodes, each PN device has a colored background:

- Green – The watchdog factor is large enough
- Red – The watchdog factor is too low and will lead to a compile error in the TIA Portal

This information is also shown as colored circle for the overall system.

IO systems with MRP

Non-redundant systems do not have this background or the circle indicator. Instead, there is an indication when MRP is enabled on one of the interfaces. The indication appears if the interface of the controller or of one of the PN devices is configured as MRP manager or client.

This "MRP in use" display also has a colored indicator:

- Green – The resulting watchdog time of all devices is at least 200 ms
- Yellow – The resulting watchdog time is less than the recommended 200 ms. The yellow color serves as a warning and not as an error, since values under 200 ms can also result in a stable MRP system but this is the user's responsibility.

The rule is not only applied on MRP clients, because the topology is not known and it is assumed that all PN IO devices would be affected by a ring reconfiguration.

IO systems without MRP

PN devices that are assigned to IO controllers without MRP function are also listed. If necessary, the watchdog time can be adjusted for these devices also. This is advisable in configurations in which WLAN is used as communication medium, for example.

Value ranges

The reconfiguration time used for calculating the factors is 200 ms when the tool is started and can be individually adapted in a range from 6 ms to 1920 ms, enabled and applied. The application of a changed value always only affects the devices selected at that time.

The tool limits the factors in the application to the permissible values in the range from 3 to 255.

1.3. Components used

This Add-In was created for these software components:

Table 1-1

Component	Note
TIA Portal V16	R/H Add-In Version 1.1.1
TIA Portal V17	R/H Add-In Version 1.2 (last version)
TIA Portal V18	PN Watchdog Add-In Version 1.3 (first version)
TIA Portal V19	PN Watchdog Add-In Version 1.4

2. Use

2.1. Installation

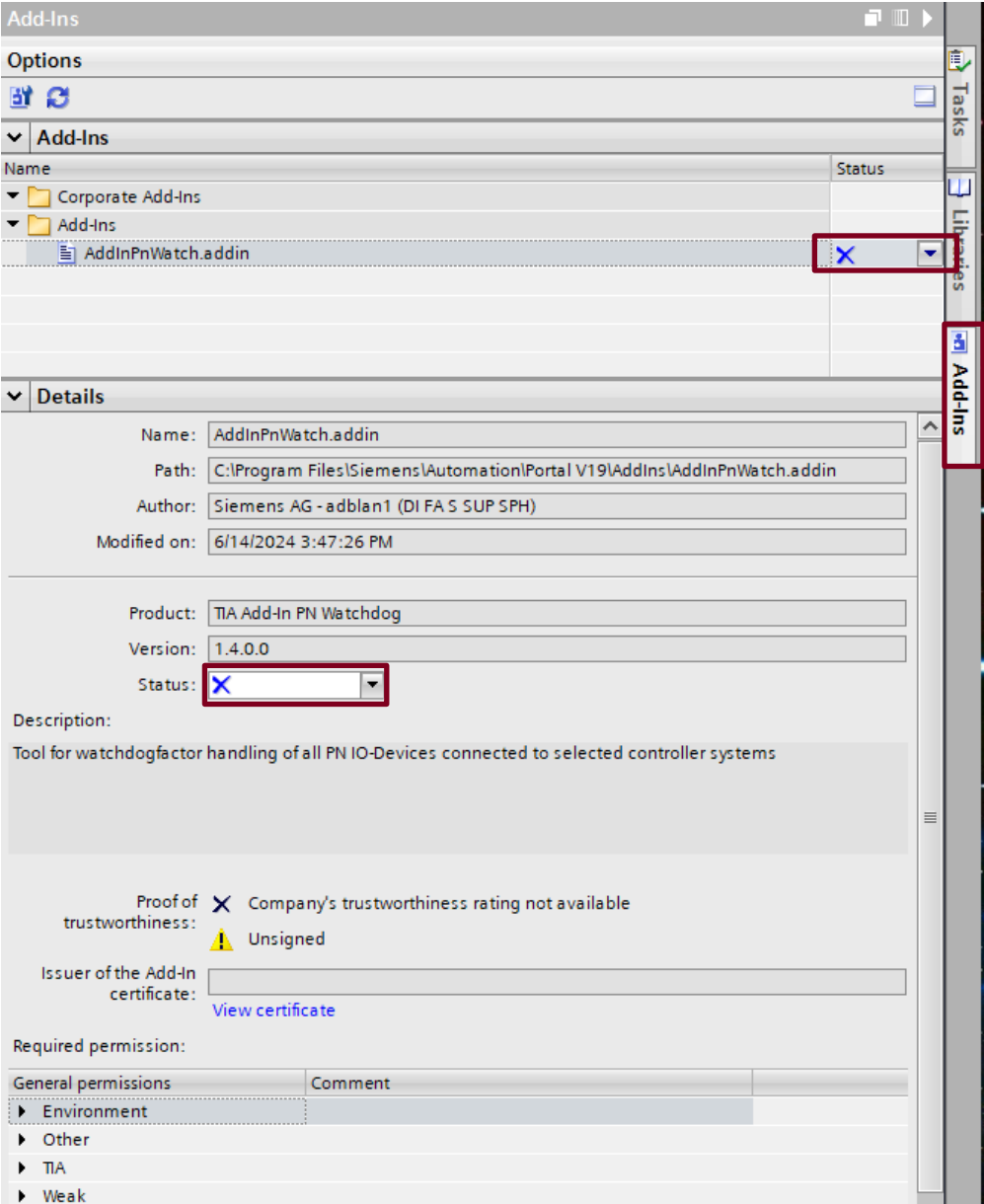
The Add-In consists of the "AddInPnWatch.addin" file. In order to access the functions of the Add-In in the TIA Portal, the "AddIns" folder must be added to the TIA Portal program folder, if it is not yet present:

.\Program Files\Siemens\Automation\Portal V1x\Add-Ins

The .addin file is inserted into this folder.

Activation of the Add-Ins takes place via a tab (see Figure 2-1) on the right-hand side of the project view.

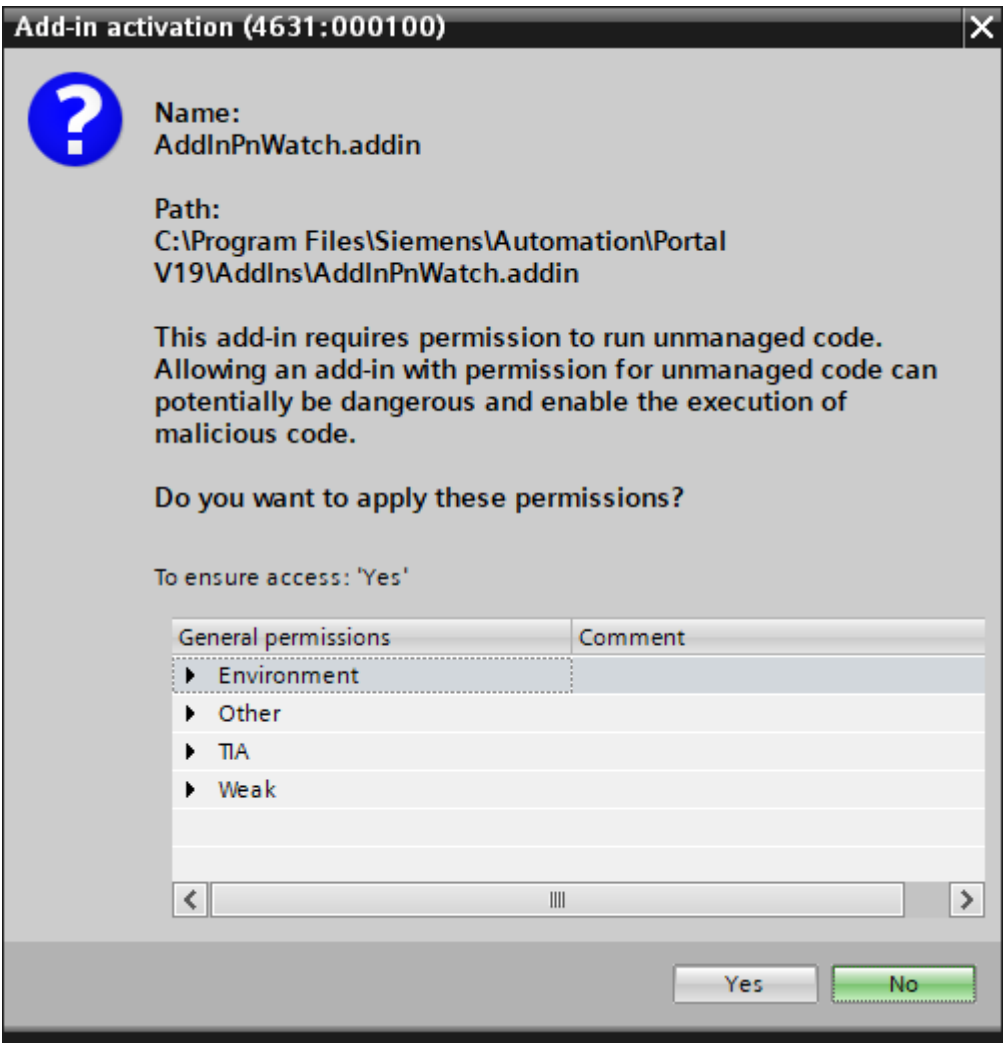
Figure 2-1: TIA Portal Add-In Tab



On the Add-Ins tab, you search for the "AddInPnWatch.addin" file in the "AddIns" folder and activate this Add-In using the directly adjacent drop-down list or in the details of the Add-In.

You are then prompted to accept the required permissions for execution of the Add-In. After confirmation, the Add-In is activated and can be used directly.

Figure 2-2: Activation dialog of Add-In authorizations



To use the Add-In, the user must belong to the "Siemens TIA Openness" user group.

You can find more information on setting up the Openness function in the manual "SIMATIC Openness: Automating creation of projects" <https://support.industry.siemens.com/cs/ww/en/view/109477163>

2.2. Operation

The Add-In is addressed via the shortcut menu function (right mouse button) in the project tree or in the network/topology view.

In the project tree, this is possible at the project, group and system level and, in redundant systems, also at the PLC level. Multiple selection is supported.

Figure 2-3: Entry points for the Add-In in the project tree

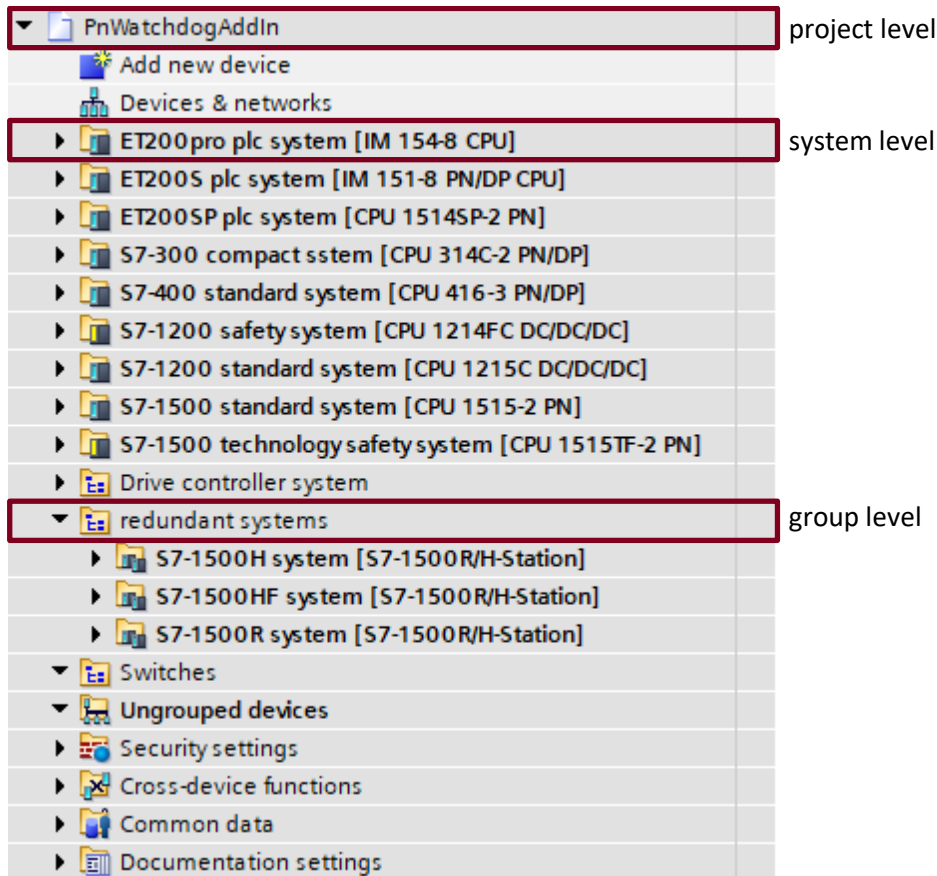
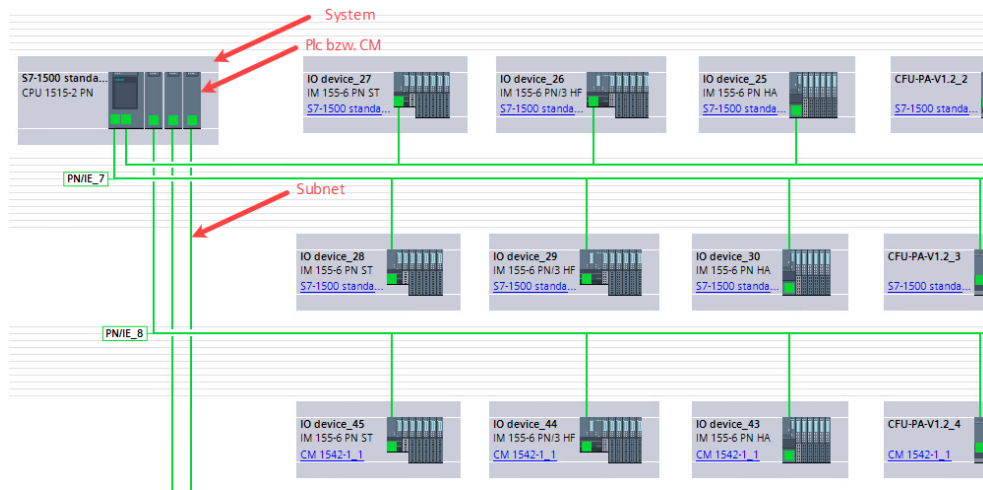


Figure 2-4

In the network view or topology view, the menu can be opened via systems and subnets. Multiple systems or subnets () can also be selected at this point.

Selection of individual PN IO devices or IO systems () is not supported or results in the device list of the tool remaining empty.

Figure 2-5: Entry points for the Add-In in the network view



All PN IO devices in the selection are searched for and listed in the tool.

The "PN Watchdog" entry is located in the bottom part of the shortcut menu with the "Show all PN Devices" function.

The "Show all PN IO-Devices" function opens a new window that reads in and lists all PN IO devices. The following information is provided for each device:

Selected (Select)

PROFINET Device Number (PN no.)

PROFINET Device Name (PN name)

PROFINET update time of the device (Update Time in ms)

Watchdog factor of the device (Watchdog factor)

Watchdog time of the device (Watchdog time in ms)

MRP role

This function allows you to make all selected systems and PN IO devices visible and to explicitly specify the PN IO devices to be updated or, if desired, to set your own values for the watchdog factor.

To maintain clarity in the case of large projects, the systems and their PROFINET IO interfaces are grouped. The top group level shows the name of the system or the name of the PLC and the number of PN IO devices assigned to the system.

The second group level is formed by the interfaces of the system that are enabled as PN IO controllers and have assigned PN IO devices.

The R/H/HF systems are limited to the X1 interface. In R1 configurations, note that only the left IM of the R1 stations is listed. Changes to the IMs in the Add-In are automatically mirrored on the right-hand IMs in the TIA Portal.

In all other systems, internal interfaces and the interfaces of the communication modules are shown.

The grouping makes it possible to set appropriate factors for each IO system and also for a large number of PN IO devices with just a few steps.

Figure 2-6: List view of the Add-In

Tool to Set/Reset the correct watchdog factor of PN IO-Devices

Network Reconfiguration Time

user value: 200 ms **Activate** active value: 200 ms

Update **Reset**

PN Nr.	PN name	Update time in ms	Watchdog factor	Watchdog time in ms	MRP Role
☐	S7-1200 station_1 (S7-1200 standard system) 10 device(s)				
☐	PROFINET interface_1 (internal.X1) 10 device(s)				
☐	S7-1500/ET200MP station_1 (S7-1500 standard system) 50 device(s)				MRP in use
☐	PROFINET interface_1 (internal.X1) 10 device(s)				
☐	PROFINET interface_2 (internal.X2) 10 device(s)				
☐	PROFINET interface_1 (CM 1542-1_1.X1) 10 device(s)				MRP in use
☐	PROFINET interface_1 (CM 1542-1_2.X1) 10 device(s)				
☐	PROFINET interface_1 (CM 1542-1_3.X1) 10 device(s)				MRP in use
☐	S7-1200 station_2 (S7-1200 safety system) 10 device(s)				
☐	PROFINET interface_1 (internal.X1) 10 device(s)				
☐	S7-1500 station_2 (S7-1500 safety system) 5 device(s)				

Collapse all **Transfer & Close**

When the tool starts, the reconfiguration time of 200 ms recommended for MRP rings is used as a basis for calculating the factors. This value can be individually adapted and applied.

The "active value" output field shows the reconfiguration time value that is currently being used for calculations.

The reconfiguration time value can be selected in a range from 6 to 1920 ms.

The "Update" button sets the correct watchdog factor for all selected devices as a function of the selected reconfiguration time of the network. The factor for a PN IO device is limited to a range from 3 to 255. This limitation is due to the values that can be set in the TIA Portal.

The "Reset" button sets the watchdog factor for all selected devices to the default start value 3.

If you close the tool using the red X button in the top right corner, all changes are discarded and the tool is immediately closed. The project data is not updated.

"Transfer & Close" applies the changes to the TIA Portal project and closes the Add-In. The tool cannot be closed while saving is in progress. The progress of the operation is indicated by two progress bars.

3. Appendix

3.1. Service and support

SiePortal

The integrated platform for product selection, purchasing and support - and connection of Industry Mall and Online support. The SiePortal home page replaces the previous home pages of the Industry Mall and the Online Support Portal (SIOS) and combines them.

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In Products & Services, you can find all our offerings as previously available in Mall Catalog.
- **Support**
In Support, you can find all information helpful for resolving technical issues with our products.
- **mySieportal**
mySiePortal collects all your personal data and processes, from your account to current orders, service requests and more. You can only see the full range of functions here after you have logged in.

You can access SiePortal via this address: sieportal.siemens.com

Industry Online Support

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Please send queries to Technical Support via Web form: support.industry.siemens.com/cs/my/src

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Industry Online Support app

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3.2. Industry Mall



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mall.industry.siemens.com

3.3. Links and literature

Table 3-1

No.	Topic
11\	Siemens Industry Online Support https://support.industry.siemens.com
12\	Link to this entry page of this application example https://support.industry.siemens.com/cs/ww/en/view/109769093
13\	SIMATIC Openness: Automating creation of projects https://support.industry.siemens.com/cs/ww/en/view/109477163

3.4. Change documentation

Table 3-2

Version	Date	Modifications
V1.0	06/2019	First edition
V1.1	01/2020	Add-In for TIA Portal V15.1 was removed Expanded to include the Add-In for TIA Portal V16 in Version 1.1
V1.1.1	05/2021	Recertification of the Add-In for TIA Portal V16 and adaptation of the Add-In layout
V1.2	05/2021	Release of the Add-In for TIA Portal V17 Addition of PLC 1518HF Extension of the Add-In to enable setting of the reconfiguration time of the network in the list view
V1.3	02/2023	Switch from the R/H Add-In to the PN Watchdog Add-In Addition of all PN IO controllers that can be configured in the TIA Portal
V1.4	06/2024	Release of the PN Watchdog Add-In for TIA Portal V19